

CL'Axioms and rules

- I 1. $IM \supseteq_1 M$
 2. $KMN \supseteq_1 M$
 3. $SMNL \supseteq_1 ML(NL)$
- II 1. $M = M$
 2. $\frac{M = N}{N = M}$
 3. $\frac{M = N, N = L}{M = L}$
 4. $\frac{M = M'}{ZM = ZM'}, \frac{M = M'}{MZ = M'Z}$
- III 1. $M \supseteq M$
 2. $\frac{M \supseteq N, N \supseteq L}{M \supseteq L}$
 3. $\frac{M \supseteq M'}{ZM \supseteq ZM'}, \frac{M \supseteq M'}{MZ \supseteq M'Z}$
 4. $\frac{M \supseteq M'}{M = M'}$
- IV 1. $M \supseteq_1 M$
 2. $\frac{M \supseteq_1 M'}{ZM \supseteq_1 ZM'}, \frac{M \supseteq_1 M'}{MZ \supseteq_1 M'Z}$
 3. $\frac{M \supseteq_1 M'}{M \supseteq M'}$
 4. $\frac{M \supseteq_1 M'}{\underline{M} \supseteq_1 \underline{M'}}$
- V 1. $M \approx M$
 2. $\frac{M \approx N}{N \approx M}$
 3. $\frac{M \approx N, N \approx L}{M \approx L}$
 4. $\frac{M \approx M'}{ZM \approx ZM'}, \frac{M \approx M'}{MZ \approx M'Z}$
 5. $M \approx \underline{M}$
- VI $\underline{MN} \supseteq_1 \underline{M} \mid \underline{N}$

In the above M, M', N, L and Z denote arbitrary terms except in IV 4, V 5 and VI where M, M' denote simple terms.

Appendix II

The Church-Rosser theorem for the λ -calculus à la Martin-Löf

This appendix contains a proof of the Church-Rosser theorem recently discovered by Martin-Löf [1971].

This proof is strikingly simple compared to those mentioned in 1.2.18.

The idea of the proof arose from cut elimination properties of certain formal systems. In fact the Church-Rosser theorem is a kind of cut elimination theorem, the transitivity of $=$ in the λ -calculus corresponding to the cut.

The trick is to define a relation $\geq_1$ between terms in such a way that

- 1) The transitive closure of $\geq_1$ is the (classical) reduction relation ($\geq$).
- 2) If $M_1 \geq_1 M_2$, $M_1 \geq_1 M_3$, then there exists a term M_4 such that $M_2 \geq_1 M_4$ and $M_3 \geq_1 M_4$.

From 1) and 2) the analogue of 2) for $\geq$ can be derived.

From this the Church-Rosser theorem easily follows.

Definition 1.

λ' is a theory formulated in the following language:

Alphabet: a,b,c,...	variables
$\lambda, (, ,)$	improper symbols
$=$	equality
$\geq$	reduction
$\geq_1$	one step reduction

Terms: The terms are defined as in the λ -calculus (1.1.1).

Formulas: If M, N are terms, then $M = N$, $M \geq N$ and $M \geq_1 N$ are formulas.

As in 1.1.2 we define $BV(M)$, $FV(M)$ and $[x/N]M$.

Definition 2.

λ' is defined by the following axioms and rules.

- I 1.
$$\frac{M \geq_1 M'}{(\lambda x M) \geq_1 \lambda y [x/y] M'} \quad \text{if } y \notin FV(M')$$
2.
$$\frac{M \geq_1 M', N \geq_1 N'}{(\lambda x M) N \geq_1 [x/N'] M'} \quad \text{if } BV(M') \cap FV(N') = \emptyset$$
- II. 1. $M \geq_1 M$
2.
$$\frac{M \geq_1 M', N \geq_1 N'}{MN \geq_1 M'N'}$$
3.
$$\frac{M \geq_1 M'}{\lambda x M \geq_1 \lambda x M'}$$
4.
$$\frac{M \geq_1 M'}{M \geq M'}$$
- III 1.
$$\frac{M \geq N, N \geq L}{M \geq L}$$
2.
$$\frac{M \geq M'}{ZM \geq ZM'}, \frac{M \geq M'}{MZ \geq M'Z}, \frac{M \geq M'}{\lambda x M \geq \lambda x M'}$$
3.
$$\frac{M \geq M'}{M = M'}$$
- IV 1.
$$\frac{M = N}{N = M}$$
2.
$$\frac{M = N, N = L}{M = L}$$
3.
$$\frac{M = M'}{ZM = ZM'}, \frac{M = M'}{MZ = M'Z}, \frac{M = M'}{\lambda x M = \lambda x M'}$$

In the above M, M', N, L and Z denote arbitrary terms and x, y denote arbitrary variables.

Lemma 3.

$$\lambda' \vdash M \geq N \iff \exists N_1 \dots N_k \quad \lambda' \vdash M \equiv N_1 \geq_1 \dots \geq_1 N_k \equiv N.$$

Proof.

$\Leftarrow$: Immediate.

$\Rightarrow$: Induction on the length of proof of $\lambda' \vdash M \geq N$. □

Lemma 4.

$$\lambda' \vdash M \geq_1 N \Rightarrow \lambda \vdash M \geq N$$

$$\lambda' \vdash M \geq N \iff \lambda \vdash M \geq N$$

$$\lambda' \vdash M = N \iff \lambda \vdash M = N$$

Proof.

In all cases induction on the length of proof. □

Lemma 5.

If $\lambda' \vdash M \geq_1 M'$ and $\lambda' \vdash N \geq_1 N'$, then

$$\lambda' \vdash [x/N]M \geq_1 [x/N']M'.$$

Proof.

Induction on the length of proof of $M \geq_1 M'$ using the sublemma:

$$\text{If } x \neq y \text{ then } [x/N_1]([y/N_2]M) \equiv [y/[x/N_1]N_2][x/N_1]M.$$

The proof of the sublemma proceeds by induction on the structure of M . □

Lemma 6.

1) If $\lambda' \vdash \lambda x M \geq_1 N$, then there exists a term M' such that

$$\lambda' \vdash M \geq_1 M' \quad \text{and} \quad N \equiv \lambda x M' \quad \text{or} \quad N \equiv \lambda y [x/y] M' \quad \text{with } y \notin \text{FV}(M').$$

2) If $\lambda' \vdash M_1 M_2 \geq_1 N$, then there exist M'_1, M'_2 such that

$$\lambda' \vdash M_1 \geq_1 M'_1 \quad \text{and} \quad N \equiv M'_1 M'_2 \quad \text{or} \quad M_1 = (\lambda x M'_1) \quad \text{and} \quad N \equiv [x/M'_2] M'_1$$

$$\text{where } \lambda' \vdash M'_1 \geq_1 M''_1 \quad \text{and} \quad \lambda' \vdash M_2 \geq_1 M'_2.$$

Proof.

Induction on the length of proof. □

Lemma 7.

If $\lambda' \vdash M_1 \geq_1 M_2$ and $\lambda' \vdash M_1 \geq_1 M_3$, then there exists a term M_4 such that $\lambda' \vdash M_2 \geq_1 M_4$ and $\lambda' \vdash M_3 \geq_1 M_4$.

Proof.

Induction on the sum of lengths of proof of $M_1 \geq_1 M_2$ and $M_1 \geq_1 M_3$.

case 1. $M_1 \geq_1 M_2$ is an axiom. Then $M_1 \equiv M_2$ and we can take

$$M_4 \equiv M_3.$$

case 2. $M_1 \geq_1 M_2$ is $\lambda x M \geq_1 \lambda y [x/y] M'$ where $y \notin FV(M')$ and is a direct consequence of $M \geq_1 M'$.

By lemma 6 it follows that

$$M_3 \equiv \lambda y' [x/y'] M'' \text{ where } \lambda \vdash M \geq_1 M'' \text{ and } y' \notin FV(M'') \text{ or } y' = x.$$

By the induction hypothesis there exists a M''' such that

$$\lambda' \vdash M' \geq_1 M''' \text{ and } \lambda' \vdash M'' \geq_1 M'''. \text{ Then we can take}$$

$$M_4 \equiv \lambda y'' [x/y''] M''' \text{ with } y'' \notin FV(M''').$$

case 3. $M_1 \geq_1 M_2$ is $\lambda x M \geq_1 \lambda x M'$ and is a direct consequence of $M \geq_1 M'$. Analogously to case 2 we can find the required term M_4 .

case 4. $M_1 \geq_1 M_2$ is $(\lambda x M) N \geq_1 [x/N'] M'$ and is a direct consequence of $M \geq_1 M'$, $N \geq_1 N'$.

By lemma 6 we can distinguish the following subcases.

subcase 4.1. $M_3 \equiv (\lambda y [x/y] M'') N''$, where $\lambda' \vdash M \geq_1 M''$,
 $\lambda' \vdash N \geq_1 N''$.

By the induction hypothesis there exist terms M''' , N''' such that $\lambda' \vdash M' \geq_1 M'''$;

$\lambda' \vdash M'' \geq_1 M'''$, $\lambda' \vdash N' \geq_1 N'''$ and
 $\lambda' \vdash N'' \geq_1 N'''$.

Then by lemma 5 we can take $M_4 \equiv [x/N''']M'''$.

subcase 4.2. $M_3 \equiv [x/N'']M''$ with $\lambda' \vdash M \geq_1 M''$,
 $\lambda' \vdash N \geq_1 N''$.

By the induction hypothesis there exist
terms M''' , N''' such that $\lambda' \vdash M' \geq_1 M'''$ etc.

Then by lemma 5 we can take $M_4 \equiv [x/N''']M'''$.

case 5. $M_1 \geq_1 M_2$ is $MN \geq_1 M'N'$ and is a direct consequence of
 $M \geq_1 M'$, $N \geq_1 N'$.

By lemma 6 we can distinguish the following subcases.

subcase 5.1. $M_3 \equiv M''N''$ with $\lambda' \vdash M \geq_1 M''$, $\lambda' \vdash N \geq_1 N''$.

By the induction hypothesis there exist
terms M''' , N''' such that $\lambda' \vdash M' \geq_1 M'''$ etc.

Then we can take $M_4 \equiv M'''N'''$.

subcase 5.2. $M_1 \geq M_3$ is $(\lambda x M_1)N \geq [x/N'']M'_1$ and is a
direct consequence of $M_1 \geq_1 M'_1$, $N \geq_1 N''$.

This case is analogous to subcase 4.1. $\square$

Lemma 8.

If $\lambda' \vdash M_1 \geq M_2$ and $\lambda' \vdash M_1 \geq M_3$, then there exists a term M_4
such that $\lambda' \vdash M_2 \geq M_4$ and $\lambda' \vdash M_3 \geq M_4$.

Proof.

By lemma 3 $\lambda' \vdash M_1 \geq M_2 \iff \exists N_1 \dots N_k \lambda' \vdash M_1 \equiv N_1 \geq_1 \dots \geq_1 N_k \equiv N$
and similarly for $\lambda' \vdash M_1 \geq M_3$.

By repeated use of lemma 7 (see figure 2, page 40) it follows
that the conclusion holds. $\square$

Lemma 9.

If $\lambda' \vdash M = N$, then there exists a term Z such that
 $\lambda' \vdash M \geq Z$ and $\lambda' \vdash N \geq Z$.

Proof.

Induction on the length of proof of $M = N$, using lemma 8 in the case of transitivity of $=$ □

Theorem 10. (Church-Rosser theorem)

If $\lambda \vdash M = N$, then there exists a term Z such that
 $\lambda \vdash M \geq Z$ and $\lambda \vdash N \geq Z$.

Proof.

This follows immediately from lemma 9 and lemma 4. □

Remark.

In the same way we can prove the Church-Rosser theorem for $\lambda + \text{ext}$ by adding to λ' the rule

$$\frac{M \geq_1 M'}{\lambda x(Mx) \geq_1 M'}$$